

XR School Implemented Simulation Quality Report

Audit date: 2026-08-01

Scope: 41 released simulations

Portfolio average: 79.3/100

Evidence maturity: 41 internal QA; 0 device verified; 0 classroom verified

Audit position: Released means publicly launchable. Released does not mean school-validated, Quest-verified, classroom-verified, or proven to improve learning outcomes.

Executive summary

The released portfolio contains 41 canonical simulations. Its evidence-backed product-indicator average is 79.3/100: 1 pilot candidates, 40 promising internal-QA classes, 0 needing focused improvement, and 0 requiring rebuild before pilot.

Repository evidence records 283 narration cues, 283 packaged narration clips, 0 missing narration files, and 108 declared assets. These are implementation indicators, not learner-outcome measurements.

Quest and classroom evidence are absent: no signed physical-device acceptance runs or controlled classroom studies are represented in this audit. Every class therefore remains at internal QA evidence maturity.

Ranked portfolio

Rank | Simulation | Canonical slug | Score | Band | Evidence maturity

1	Soluble and Insoluble Substances Lab	c5-ch07-a03-soluble-and-insoluble-substances	85	Pilot candidate	internalQA
2	The Secret Life of Fungi	c8-ch02-a03-fungi-and-its-development	84	Promising	internal QA internalQA
3	Plant Pollination & Growth Cycle	pollination	83	Promising	internal QA internalQA
4	Sorting Materials According to Their Shape	c6-ch04-a01-sorting-materials-according-to-their-shape	82	Promising	internal QA internalQA
5	Sources of Vitamins and Their Deficiencies	c6-ch02-a04-the-sources-of-vitamins-and-their-deficiencies	82	Promising	internal QA internalQA
6	Test the Presence of Lipids	c6-ch02-a03-test-the-presence-of-lipids	82	Promising	internal QA internalQA
7	The Sources of Minerals in Food	c6-ch02-a05-the-sources-of-minerals-in-food	82	Promising	internal QA internalQA
8	What Floats, What Sinks?	c5-ch07-a01-a-concept-about-what-floats-what-sinks	82	Promising	internal QA internalQA
9	Harvesting, Threshing and Storage of Crops	c8-ch01-a05-harvesting-threshing-and-storage-of-crops	81	Promising	internal QA internalQA
10	Irrigation Methods: Every Root Gets Water	c8-ch01-a03-irrigation-methods	81	Promising	internal QA internalQA
11	Shearing and Scouring of Wool	c7-ch03-a01-shearing-and-scouring-of-wool	81	Promising	internal QA internalQA
12	Solar System: Gravity's Orchestra	c8-10-science-solar-system	81	Promising	internal QA internalQA
13	Spinning and Rolling of Wool	c7-ch03-a03-spinning-and-rolling-of-wool	81	Promising	internal QA internalQA
14	States of Matter Particle Lab	c9-ch01-a02-states-of-matter	81	Promising	internal QA internalQA
15	The Invisible Invader: Break the Viral Chain	c8-ch02-a02-virus-introduction-spreading-and-its-effects	81	Promising	internal QA internalQA
16	Electric Circuits & Resistance (Ohm's Law)	circuit	80	Promising	internal QA internalQA
17	A Step Well Structure	c5-ch06-a02-a-step-well-structure	79	Promising	internal QA internalQA
18	A Visit of an Ancient Fort	c5-ch10-a01-a-visit-of-ancient-fort	79	Promising	internal QA internalQA
19	Camp in the Snow	c5-ch09-a03-camp-in-the-snow	79	Promising	internal QA internalQA
20	Cotton Farming	c6-ch03-a01-cotton-farming	79	Promising	internal QA internalQA
21	Dead Sea: Salt Water and Its Effects	c5-ch07-a02-dead-sea-salt-water-and-its-effects	79	Promising	internal QA internalQA
22	Diagnosis of Malaria	c5-ch08-a01-diagnosis-of-malaria	79	Promising	internal QA internalQA
23	Food Spoilage	c5-ch04-a01-food-spoilage	79	Promising	internal QA internalQA
24	Life Cycle of the Mosquito	c5-ch08-a02-life-cycle-of-the-mosquito	79	Promising	internal QA internalQA
25	Milk Spoilage	c5-ch04-a02-milk-spoilage	79	Promising	internal QA internalQA
26	Pitcher Plant - The Insect Hunter	c5-ch05-a01-pitcher-plant-the-insect-hunter	79	Promising	internal QA internalQA
27	River Crossing Adventure	c5-ch09-a01-river-crossing-adventure	79	Promising	internal QA internalQA
28	Rock Climbing	c5-ch09-a02-rock-climbing	79	Promising	internal QA internalQA
29	Seed Dispersal	c5-ch05-a02-seed-dispersal	79	Promising	internal QA internalQA
30	Snow Mountain Climbing	c5-ch09-a04-snow-mountain-climbing	79	Promising	internal QA internalQA
31	The Making of Aam Papad	c5-ch04-a03-the-making-of-aam-papad	79	Promising	internal QA internalQA
32	The Process of Cotton Ginning	c6-ch03-a02-the-process-of-cotton-ginning	79	Promising	internal QA internalQA
33	The Storage of Rainwater	c5-ch06-a01-the-storage-of-rainwater	79	Promising	internal QA internalQA
34	Preposition Adventure	c2-english-ch01-prepositions	76	Promising	internal QA internalQA
35	Sources of Food Sorting Lab	c6-ch01-a01-sources-of-food	76	Promising	internal QA internalQA
36	Acids, Bases & Neutralisation	c10-ch02-a01-introduction-to-acids-and-bases-and-litmus-test	75	Promising	internal QA internalQA
37	Introduction to Money	c1-math-ch01-introduction-to-money	75	Promising	internal QA internalQA
38	Introduction to the Digestive System	c5-ch03-a02-introduction-of-digestive-system	75	Promising	internal QA internalQA
39	The Effects of Force on an Object's Motion and Shape	c8-ch10-a02-the-effects-of-force-on-object-s-motion-and-shape	75	Promising	internal QA internalQA
40	Colour Adventure	c1-art-a01-learning-of-colours	74	Promising	internal QA internalQA
41	The Breathing Process in Human	c7-ch10-a02-the-breathing-process-in-human	73	Promising	internal QA internalQA

Portfolio priorities

Priorities are derived from the three lowest average rubric attainment ratios across the complete 41-card dataset.

1. Deployment readiness: portfolio mean 3.0/5. Address the card-level evidence gaps and next actions before raising evidence maturity.
2. Visual and asset quality: portfolio mean 9.5/15. Address the card-level evidence gaps and next actions before raising evidence maturity.
3. Narration and sound: portfolio mean 7.2/10. Address the card-level evidence gaps and next actions before raising evidence maturity.
4. Physical-device acceptance: run the documented Quest comfort, controller, narration, cleanup, and performance checks; current signed run count is 0.
5. Classroom evidence: collect teacher workflow and learner-comprehension evidence without converting internal QA scores into outcome claims; current study count is 0.

Quality cards

Soluble and Insoluble Substances Lab - 85/100

Canonical slug: c5-ch07-a03-soluble-and-insoluble-substances

Route: /simulations/c5-ch07-a03-soluble-and-insoluble-substances

Publication status: released

Evidence maturity: internalQA

Band: Pilot candidate

Audience: Class 5 - Environmental Science, Science

Soluble and Insoluble Substances Lab is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 19 | 20

Content / scientific integrity | 15 | 15

Learner interactivity | 15 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Secret Life of Fungi - 84/100

Canonical slug: c8-ch02-a03-fungi-and-its-development

Route: /simulations/c8-ch02-a03-fungi-and-its-development

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 8 - Biology, Science

The Secret Life of Fungi is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 18 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 11 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 7 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Plant Pollination & Growth Cycle - 83/100

Canonical slug: pollination

Route: /simulations/pollination

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Classes 6-10 - Biology, Environmental Science

A treatment-and-control investigation that links flower structure, pollination, fertilisation, seed formation, and germination through evidence-gated stages.

Dimension | Score | Maximum

Educational effectiveness | 18 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 11 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
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Gaps and risks

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Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Sorting Materials According to Their Shape - 82/100

Canonical slug: c6-ch04-a01-sorting-materials-according-to-their-shape

Route: /simulations/c6-ch04-a01-sorting-materials-according-to-their-shape

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science

Sorting Materials According to Their Shape is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 18 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

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Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Sources of Vitamins and Their Deficiencies - 82/100

Canonical slug: c6-ch02-a04-the-sources-of-vitamins-and-their-deficiencies

Route: /simulations/c6-ch02-a04-the-sources-of-vitamins-and-their-deficiencies

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science, Biology

Sources of Vitamins and Their Deficiencies is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

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Educational effectiveness | 18 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 14 | 15

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Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Test the Presence of Lipids - 82/100

Canonical slug: c6-ch02-a03-test-the-presence-of-lipids

Route: /simulations/c6-ch02-a03-test-the-presence-of-lipids

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science, Biology

Test the Presence of Lipids is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 18 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

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- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Sources of Minerals in Food - 82/100

Canonical slug: c6-ch02-a05-the-sources-of-minerals-in-food

Route: /simulations/c6-ch02-a05-the-sources-of-minerals-in-food

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science, Biology

The Sources of Minerals in Food is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 18 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 14 | 15

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Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

What Floats, What Sinks? - 82/100

Canonical slug: c5-ch07-a01-a-concept-about-what-floats-what-sinks

Route: /simulations/c5-ch07-a01-a-concept-about-what-floats-what-sinks

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science, Science

What Floats, What Sinks? is now a canonical interactive class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 18 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Harvesting, Threshing and Storage of Crops - 81/100

Canonical slug: c8-ch01-a05-harvesting-threshing-and-storage-of-crops

Route: /simulations/c8-ch01-a05-harvesting-threshing-and-storage-of-crops

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 8 - Biology, Science

Harvesting, Threshing and Storage of Crops is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 11 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Irrigation Methods: Every Root Gets Water - 81/100

Canonical slug: c8-ch01-a03-irrigation-methods

Route: /simulations/c8-ch01-a03-irrigation-methods

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 8 - Biology, Science

Irrigation Methods: Every Root Gets Water is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 11 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Shearing and Scouring of Wool - 81/100

Canonical slug: c7-ch03-a01-shearing-and-scouring-of-wool

Route: /simulations/c7-ch03-a01-shearing-and-scouring-of-wool

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 7 - Biology, Science

Shearing and Scouring of Wool is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 11 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 10 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 10 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Solar System: Gravity's Orchestra - 81/100

Canonical slug: c8-10-science-solar-system

Route: /simulations/c8-10-science-solar-system

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Classes 8-10 - Science, Physics, Geography

A concept-rich mission addressing orbit, temperature, scale, comet behavior, and common astronomy misconceptions.

Dimension | Score | Maximum

Educational effectiveness | 19 | 20

Content / scientific integrity | 15 | 15

Learner interactivity | 15 | 15

Visual and asset quality | 10 | 15

Narration and sound | 4 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 7 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Spinning and Rolling of Wool - 81/100

Canonical slug: c7-ch03-a03-spinning-and-rolling-of-wool

Route: /simulations/c7-ch03-a03-spinning-and-rolling-of-wool

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 7 - Biology, Science

Spinning and Rolling of Wool is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 11 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 10 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 10 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

States of Matter Particle Lab - 81/100

Canonical slug: c9-ch01-a02-states-of-matter

Route: /simulations/c9-ch01-a02-states-of-matter

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 9 - Chemistry, Physics

A particle-model laboratory that makes spacing, motion, attraction, heating, and phase changes observable without presenting the model as molecular footage.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 11 | 15

Narration and sound | 7 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 4 declared stages connect the curriculum objective to observable learner actions and evidence.
- 3 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
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Gaps and risks

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- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Invisible Invader: Break the Viral Chain - 81/100

Canonical slug: c8-ch02-a02-virus-introduction-spreading-and-its-effects

Route: /simulations/c8-ch02-a02-virus-introduction-spreading-and-its-effects

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 8 - Biology, Science

The Invisible Invader: Break the Viral Chain is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 11 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
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Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Electric Circuits & Resistance (Ohm's Law) - 80/100

Canonical slug: circuit

Route: /simulations/circuit

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Classes 6-10 - Physics

A tested circuit investigation connecting component manipulation, current flow, resistance, and prediction through Ohm's law.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 10 | 15

Narration and sound | 7 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

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Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Asset richness and visual clarity still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

A Step Well Structure - 79/100

Canonical slug: c5-ch06-a02-a-step-well-structure

Route: /simulations/c5-ch06-a02-a-step-well-structure

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

A Step Well Structure is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

A Visit of an Ancient Fort - 79/100

Canonical slug: c5-ch10-a01-a-visit-of-ancient-fort

Route: /simulations/c5-ch10-a01-a-visit-of-ancient-fort

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

A Visit of an Ancient Fort is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Camp in the Snow - 79/100

Canonical slug: c5-ch09-a03-camp-in-the-snow

Route: /simulations/c5-ch09-a03-camp-in-the-snow

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Camp in the Snow is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Cotton Farming - 79/100

Canonical slug: c6-ch03-a01-cotton-farming

Route: /simulations/c6-ch03-a01-cotton-farming

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Environmental Science

Cotton Farming is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Dead Sea: Salt Water and Its Effects - 79/100

Canonical slug: c5-ch07-a02-dead-sea-salt-water-and-its-effects

Route: /simulations/c5-ch07-a02-dead-sea-salt-water-and-its-effects

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Dead Sea: Salt Water and Its Effects is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Diagnosis of Malaria - 79/100

Canonical slug: c5-ch08-a01-diagnosis-of-malaria

Route: /simulations/c5-ch08-a01-diagnosis-of-malaria

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Diagnosis of Malaria is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Food Spoilage - 79/100

Canonical slug: c5-ch04-a01-food-spoilage

Route: /simulations/c5-ch04-a01-food-spoilage

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Food Spoilage is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Life Cycle of the Mosquito - 79/100

Canonical slug: c5-ch08-a02-life-cycle-of-the-mosquito

Route: /simulations/c5-ch08-a02-life-cycle-of-the-mosquito

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Life Cycle of the Mosquito is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Milk Spoilage - 79/100

Canonical slug: c5-ch04-a02-milk-spoilage

Route: /simulations/c5-ch04-a02-milk-spoilage

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Milk Spoilage is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Pitcher Plant - The Insect Hunter - 79/100

Canonical slug: c5-ch05-a01-pitcher-plant-the-insect-hunter

Route: /simulations/c5-ch05-a01-pitcher-plant-the-insect-hunter

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Pitcher Plant - The Insect Hunter is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

River Crossing Adventure - 79/100

Canonical slug: c5-ch09-a01-river-crossing-adventure

Route: /simulations/c5-ch09-a01-river-crossing-adventure

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

River Crossing Adventure is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Rock Climbing - 79/100

Canonical slug: c5-ch09-a02-rock-climbing

Route: /simulations/c5-ch09-a02-rock-climbing

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Rock Climbing is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Seed Dispersal - 79/100

Canonical slug: c5-ch05-a02-seed-dispersal

Route: /simulations/c5-ch05-a02-seed-dispersal

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Seed Dispersal is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Snow Mountain Climbing - 79/100

Canonical slug: c5-ch09-a04-snow-mountain-climbing

Route: /simulations/c5-ch09-a04-snow-mountain-climbing

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

Snow Mountain Climbing is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Making of Aam Papad - 79/100

Canonical slug: c5-ch04-a03-the-making-of-aam-papad

Route: /simulations/c5-ch04-a03-the-making-of-aam-papad

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

The Making of Aam Papad is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Process of Cotton Ginning - 79/100

Canonical slug: c6-ch03-a02-the-process-of-cotton-ginning

Route: /simulations/c6-ch03-a02-the-process-of-cotton-ginning

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science

The Process of Cotton Ginning is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Storage of Rainwater - 79/100

Canonical slug: c5-ch06-a01-the-storage-of-rainwater

Route: /simulations/c5-ch06-a01-the-storage-of-rainwater

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science

The Storage of Rainwater is now a canonical guided class with declared stages, evidence gates, assessment, narration, an auditable visual implementation, route ownership, and release metadata.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 7 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 7 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Contributor-supplied panorama provenance remains incomplete, so visuals are capped in the audit.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Preposition Adventure - 76/100

Canonical slug: c2-english-ch01-prepositions

Route: /simulations/c2-english-ch01-prepositions

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 2 - English

A spatial language lesson in which object placement makes position words concrete and supports sentence practice.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 11 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 11 | 15

Narration and sound | 4 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Sources of Food Sorting Lab - 76/100

Canonical slug: c6-ch01-a01-sources-of-food

Route: /simulations/c6-ch01-a01-sources-of-food

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 6 - Science, Biology

A classification investigation with targeted feedback that separates plant, animal, and fungal food sources.

Dimension | Score | Maximum

Educational effectiveness | 16 | 20

Content / scientific integrity | 12 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 9 | 15

Narration and sound | 8 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 7 | 10

Deployment readiness | 3 | 5

Strengths

- 4 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 4 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Acids, Bases & Neutralisation - 75/100

Canonical slug: c10-ch02-a01-introduction-to-acids-and-bases-and-litmus-test

Route: /simulations/c10-ch02-a01-introduction-to-acids-and-bases-and-litmus-test

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 10 - Chemistry

An experiment-bench sequence connecting litmus, universal indicator, pH classification, and neutralisation.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 8 | 15

Narration and sound | 4 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 7 | 10

Deployment readiness | 3 | 5

Strengths

- 5 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 5 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Introduction to Money - 75/100

Canonical slug: c1-math-ch01-introduction-to-money

Route: /simulations/c1-math-ch01-introduction-to-money

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 1 - Mathematics

An age-appropriate progression through Indian coins, notes, value comparison, simple shopping, and memory.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 11 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 11 | 15

Narration and sound | 4 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 8 | 10

Deployment readiness | 3 | 5

Strengths

- 8 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 8 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Introduction to the Digestive System - 75/100

Canonical slug: c5-ch03-a02-introduction-of-digestive-system

Route: /simulations/c5-ch03-a02-introduction-of-digestive-system

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 5 - Environmental Science, Biology

A ten-stage pathway investigation covering organs, movement, accessory organs, absorption, recap, and healthy-habit transfer.

Dimension | Score | Maximum

Educational effectiveness | 18 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 8 | 15

Narration and sound | 4 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 7 | 10

Deployment readiness | 3 | 5

Strengths

- 10 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 10 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Effects of Force on an Object's Motion and Shape - 75/100

Canonical slug: c8-ch10-a02-the-effects-of-force-on-object-s-motion-and-shape

Route: /simulations/c8-ch10-a02-the-effects-of-force-on-object-s-motion-and-shape

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 8 - Physics

A deterministic physics investigation of starting, stopping, speeding, redirection, collision, and deformation.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 14 | 15

Learner interactivity | 14 | 15

Visual and asset quality | 8 | 15

Narration and sound | 4 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 7 | 10

Deployment readiness | 3 | 5

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Colour Adventure - 74/100

Canonical slug: c1-art-a01-learning-of-colours

Route: /simulations/c1-art-a01-learning-of-colours

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 1 - Art

An early-years recognition and memory journey with large targets, visible feedback, and a canonical release record.

Dimension | Score | Maximum

Educational effectiveness | 16 | 20

Content / scientific integrity | 11 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 12 | 15

Narration and sound | 4 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 7 | 10

Deployment readiness | 3 | 5

Strengths

- 14 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 14 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

The Breathing Process in Human - 73/100

Canonical slug: c7-ch10-a02-the-breathing-process-in-human

Route: /simulations/c7-ch10-a02-the-breathing-process-in-human

Publication status: released

Evidence maturity: internalQA

Band: Promising internal QA

Audience: Class 7 - Biology

A six-stage physiology lesson connecting airway structure, diaphragm motion, chest volume, and gas exchange.

Dimension | Score | Maximum

Educational effectiveness | 17 | 20

Content / scientific integrity | 13 | 15

Learner interactivity | 13 | 15

Visual and asset quality | 8 | 15

Narration and sound | 4 | 10

Usability, accessibility, comfort | 8 | 10

Performance and stability | 7 | 10

Deployment readiness | 3 | 5

Strengths

- 6 declared stages connect the curriculum objective to observable learner actions and evidence.
- 2 assessment prompts include misconception and transfer checks; completion is kept separate from mastery.
- 6 committed narration clips are owned by the manifest with exact captions.

Gaps and risks

- Evidence maturity remains internal QA: no signed headset acceptance run or classroom outcome study has occurred.
- Code-native visual structure has source and automated evidence; visual clarity and performance still need a representative low-end device review.
- Teacher facilitation, learner-language comprehension, and multi-session reliability still require a controlled school pilot.

Next action: Run representative browser and headset acceptance, then conduct a teacher-facilitated pilot before making learning-effect claims.

Contribution appendix

This appendix maps every PR #8 contribution to its canonical released class. Scores compare the immutable PR head with the integrated internal-QA implementation.

PR slug | Canonical slug | Before | After | Delta | Main remediation | Remaining risk

walls-tell-stories-ancient-fort-visit | c5-ch10-a01-a-visit-of-ancient-fort | 65 | 79 | +14 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

up-you-go-snow-mountain-climbing | c5-ch09-a04-snow-mountain-climbing | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

up-you-go-camp-in-snow | c5-ch09-a03-camp-in-the-snow | 51 | 79 | +28 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

up-you-go-rock-climbing | c5-ch09-a02-rock-climbing | 61 | 79 | +18 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

up-you-go-river-crossing-adventure | c5-ch09-a01-river-crossing-adventure | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

treat-for-mosquitoes-mosquito-life-cycle | c5-ch08-a02-life-cycle-of-the-mosquito | 57 | 79 | +22 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

treat-for-mosquitoes-malaria-diagnosis | c5-ch08-a01-diagnosis-of-malaria | 57 | 79 | +22 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

experiments-with-water-float-or-sink | c5-ch07-a01-a-concept-about-what-floats-what-sinks | 64 | 82 | +18 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

experiments-with-water-dead-sea-salt-water | c5-ch07-a02-dead-sea-salt-water-and-its-effects | 56 | 79 | +23 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

experiments-with-water-soluble-insoluble | c5-ch07-a03-soluble-and-insoluble-substances | 63 | 85 | +22 | Integrated the useful PR experiment as an enhancement of the existing Solubility class, avoiding a duplicate 36th simulation. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

every-drop-counts-rainwater-storage | c5-ch06-a01-the-storage-of-rainwater | 54 | 79 | +25 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

every-drop-counts-stepwell-structure | c5-ch06-a02-a-step-well-structure | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

seeds-and-seeds-seed-dispersal | c5-ch05-a02-seed-dispersal | 54 | 79 | +25 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

seeds-and-seeds-pitcher-plant | c5-ch05-a01-pitcher-plant-the-insect-hunter | 54 | 79 | +25 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

mangoes-round-the-year-aam-papad | c5-ch04-a03-the-making-of-aam-papad | 50 | 79 | +29 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

mangoes-round-the-year-milk-spoilage | c5-ch04-a02-milk-spoilage | 50 | 79 | +29 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

mangoes-round-the-year-food-spoilage | c5-ch04-a01-food-spoilage | 50 | 79 | +29 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

sorting-materials-by-shape | c6-ch04-a01-sorting-materials-according-to-their-shape | 51 | 82 | +31 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

fibre-to-fabric-cotton-farming | c6-ch03-a01-cotton-farming | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

fibre-to-fabric-cotton-ginning | c6-ch03-a02-the-process-of-cotton-ginning | 52 | 79 | +27 | Moved curriculum, narration, assessment, assets, legacy route, and release state into a validated ImplementedSimulationDefinition. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

components-of-food-mineral-sources | c6-ch02-a05-the-sources-of-minerals-in-food | 58 | 82 | +24 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

components-of-food-vitamins-deficiencies | c6-ch02-a04-the-sources-of-vitamins-and-their-deficiencies | 58 | 82 | +24 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

components-of-food-lipid-test | c6-ch02-a03-test-the-presence-of-lipids | 62 | 82 | +20 | Moved curriculum, assessment, narration, assets, legacy route, and release state into the canonical content registry. | No signed physical-headset acceptance has been run, so comfort, controller discoverability, frame rate, and listener quality remain internal-QA risks.

Rubric, evidence method, and limitations

Weighted rubric

Dimension | Weight

Educational effectiveness | 20

Content / scientific integrity | 15

Learner interactivity | 15

Visual and asset quality | 15

Narration and sound | 10

Usability, accessibility, comfort | 10

Performance and stability | 10

Deployment readiness | 5

Evidence method

Scores are computed from canonical definitions, declared stage/action/evidence/assessment counts, narration and asset manifests, focused behavior tests, production-build inclusion, browser-contract observations, and immutable PR-head evidence references.

Limitations

- Scores are product indicators, not grades for a contributor and not evidence of measured learning gain.
- Released does not mean school-validated. Publication status and evidence maturity are separate fields.
- Quest and classroom evidence are absent. Browser/build evidence cannot replace physical-headset or teacher-led acceptance.
- Scores must be recalculated after material implementation changes or new signed evidence.